Failure-case analysis

What goes wrong, how often, and what it looks like (T37). Produced by `python run.py failures`: B1, B2, M1 and M2 each sample the same 100 test rooms 64 times (6,400 samples per method; frozen configuration, seed 0). M1 and M2 start from the same draws, so sample *i* of M2 is sample *i* of M1 after latent optimization.

Every sample is classified by the hard checks it breaks: **out of room** (H1), **overlap** (H2), **door blocked** (H3), **unreachable** (H4). A valid sample is **poor** if its rule score is below 0.5, otherwise **good**.

Tables: `reports/tables/failure_causes.csv` and `failure_cases.csv`. Gallery: `reports/figures/failure_cases.png`.

1. How often each failure occurs

Share of all samples of a method. A sample can break several checks.

	B1	B2	M1	M2
Valid	9.9%	58.6%	14.4%	**67.8%**
of which good (score at least 0.5)	0.2%	37.9%	11.2%	50.1%
of which poor	9.7%	20.8%	3.1%	17.6%
Breaks H1, out of room	0.0%	0.3%	29.7%	4.0%
Breaks H2, overlap	70.9%	1.2%	56.7%	6.4%
Breaks H3, door blocked	40.9%	31.9%	34.3%	9.6%
Breaks H4, unreachable	63.3%	34.9%	45.0%	28.4%
Breaks several checks	59.8%	26.5%	53.0%	10.8%
Breaks only H4	7.1%	8.5%	1.9%	**18.5%**
Rooms with no valid layout at all	16%	0%	11%	0%

2. What the numbers say

Latent optimization repairs what it targets. From M1 to M2, on the same candidates:

Check	M1	M2	Removed
Out of room	29.7%	4.0%	87%
Overlap	56.7%	6.4%	89%
Door blocked	34.3%	9.6%	72%
Unreachable	45.0%	28.4%	37%

The first three have a term in the optimization loss. Reachability has none, because it is not differentiable; it improves only as a side effect of items moving apart.

Reachability is M2's main remaining failure. 28.4% of M2's samples leave an item unreachable, which is 88% of its invalid samples, and 18.5% of all samples fail on reachability *alone*: nothing overlaps, nothing leaves the room, the door is free, but there is no 0.6 m path to some item's front. That is more than half of M2's failures (case 4 in the gallery). Any improvement of M2 has to start here.

The raw CVAE mostly fails on overlap and on the walls. M1's decoder places each item about 0.2 m from where a good layout would have it. That is enough for two neighbouring items to overlap (56.7%) and for a wall item to stick through its wall (29.7%). The Sigmoid keeps an item's *centre* inside the room, not its footprint. These are exactly the small violations that a few gradient steps remove (cases 7 and 8).

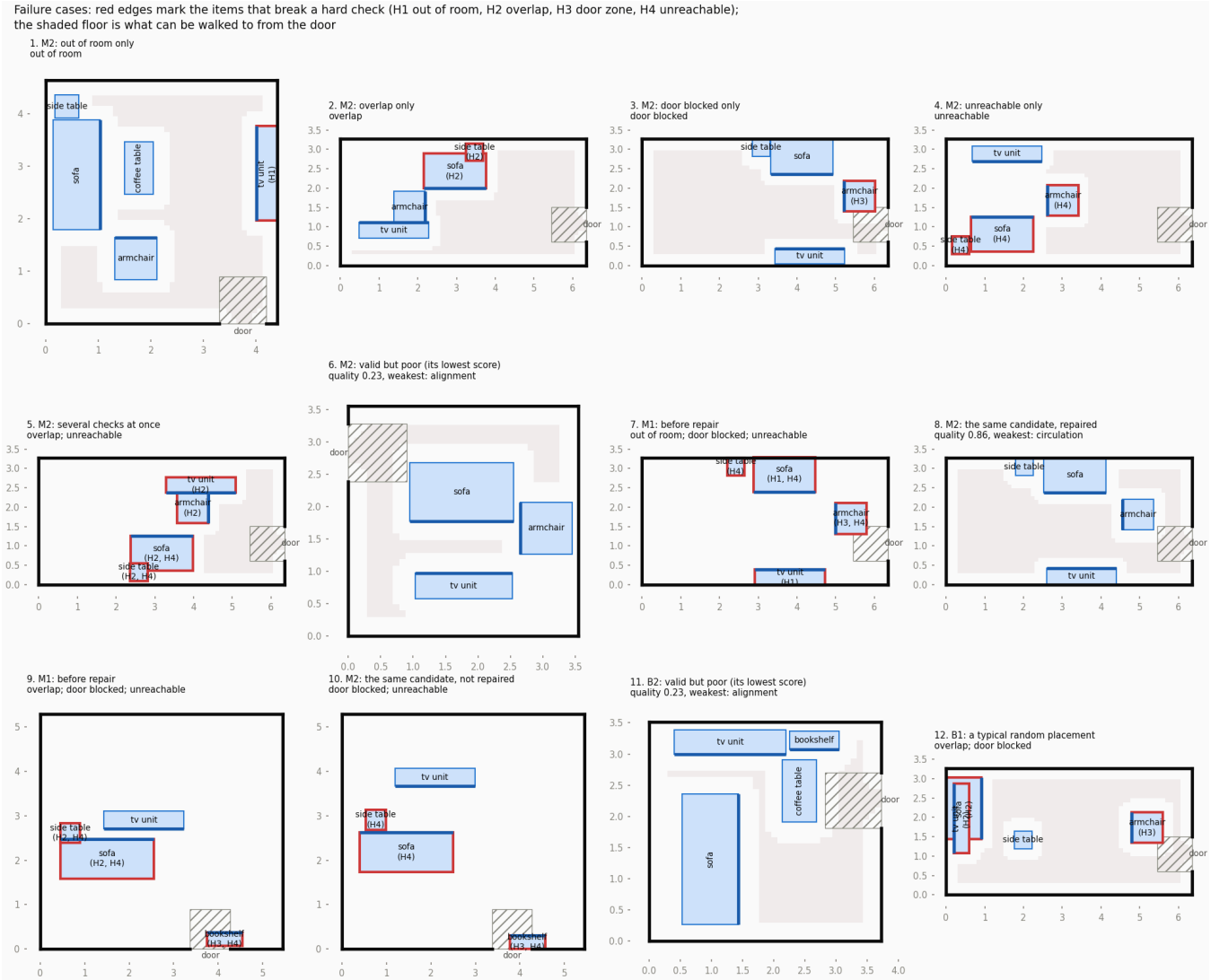
B2 fails differently. It almost never overlaps (1.2%), because it redraws an item that would. It fails on the door zone (31.9%) and on reachability (34.9%), which it never looks at.

Valid is not the same as good. A quarter of M2's valid samples score below 0.5 (17.6% of 67.8%), and a third of B2's (20.8% of 58.6%). The typical poor layout has an item that belongs against a wall standing in the open (cases 6 and 11): it breaks no hard rule and looks wrong. This is what the evaluator's ranking is for: the top 3 shown average 0.85, well above the 0.66 of all valid samples.

No collapse. No room of M2 has near-identical valid layouts (the least varied room still averages 1.05 m between them), and every room gets valid layouts. M1 leaves 11% of rooms without any valid layout, B1 16%.

3. Gallery

Red edges mark the items that break a check; the label under the item's name says which. The shaded floor is what can be walked to from the door.



Failure cases

Case	Method	What happened	Cause
1	M2	The TV unit sticks through the east wall by a few centimeters	Out of room: a small protrusion that the optimization left
2	M2	The side table overlaps the sofa's corner	Overlap. A small item next to a large one: the residual overlap is above the 0.005 m^2 tolerance
3	M2	The armchair stands in the door's clearance zone	Door blocked
4	M2	Nothing overlaps, yet the sofa, the armchair and the side table cannot be reached: the walkable floor stops short of them	Unreachable only: the passages left between the items and the walls are narrower than the 0.6 m a person needs. The loss has no term for this
5	M2	Two pairs of items still overlap, and the sofa and the side table are cut off	Several checks at once: a candidate far from any valid layout
6	M2	Valid, score 0.23: the TV unit stands away from its wall, less than a meter in front of the sofa	Poor. No hard rule is broken; alignment and the sofa-to-TV distance are
7, 8	M1, then M2	Before: the sofa and the TV unit stick out of the room and the armchair blocks the door. After: valid, score 0.86	A successful repair of the same candidate
9, 10	M1, then M2	Before: overlap, door blocked, unreachable. After: the overlap is gone, but the bookshelf still stands in the door zone and the sofa is still cut off	A failed repair. Probably the bookshelf would have to move too far: the anchor to the starting point holds every candidate back
11	B2	Valid, score 0.23: the sofa faces the door wall, and the TV unit stands on the wall to its side	Poor. Each position was copied from a different training layout; each is plausible alone, together they are not
12	B1	The sofa lies on top of the TV unit; the armchair blocks the door	Random placement

4. What follows

1. **A reachability term is the most valuable next step for M2.** More than half of its failures break nothing else. Options: a differentiable stand-in (for example a penalty on gaps narrower than 0.6 m in front of each item), or the CNN evaluator as a surrogate, with a test for the optimizer exploiting it.
2. **The door zone is repaired least of the three targeted checks** (72%). An item inside the zone has to travel up to 0.9 m, which the anchor resists. A larger door weight or more steps for such candidates would be the first thing to try.
3. **Ranking matters as much as validity** for what the user sees, because a quarter of the valid layouts are poor.
4. **The checker must stay the final authority.** Every one of these failures was caught by it, and none would have been caught reliably by the CNN alone at the tolerance boundary.